National Semiconductor

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LM397
Single General Purpose Voltage Comparator

General Description

The LM397 is a single voltage comparator with an input common mode that includes ground. The LM397 is designed to operate from a single 5V to 30V power supply or a split power supply. Its low supply current is virtually independent of the magnitude of the supply voltage.

The LM397 features an open collector output stage. This allows the connection of an external resistor at the output. The output can directly interface with TTL, CMOS and other logic levels, by tying the resistor to different voltage levels (level translator).

The LM397 is available in space saving SOT23-5 package and pin compatible to TI's TL331, single differential comparator.

Features

(T_A = 25˚C. Typical values unless otherwise specified).

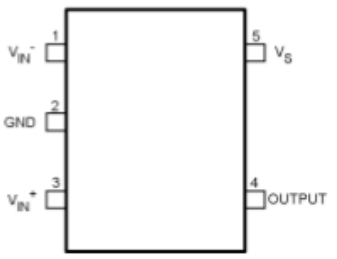
- SOT23-5 package
- Industrial operating range −40˚C to +85˚C
- Single or dual power supplies
- Wide supply voltage range 5V to 30V
- Low supply current 300μA
- Low input bias current 7nA
- Low input offset current ±1nA
- Low input offset voltage ±2mV
- Response time 440ns (50mV overdrive)
- Input common mode voltage 0 to V_S - 1.5V

Applications

- A/D converters
- Pulse, square wave generators
- Peak detector
- Industrial applications

Connection Diagram

SOT23-5



Top View

Typical Circuit

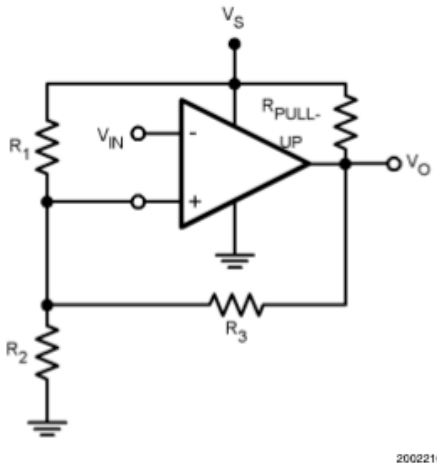


FIGURE 1. Inverting Comparator with Hysteresis

Ordering Information

Package	Part Number	Package Marking	Transport Media	NSC Drawing
5-Pin SOT-23	LM397MF	C397	1k Units Tape and Reel	MF05A
	LM397MFX		3k Units Tape and Reel	

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Absolute Maximum Ratings (Note 1)

If Military/Aerospace specified devices are required, please contact the National Semiconductor Sales Office/ Distributors for availability and specifications.

ESD Tolerance	
Human Body Model	2KV (Note 2)
Machine Model	200V (Note 3)
V_{IN} Differential	30V
Supply Voltages	30V or ±15V
Voltage at Input Pins	−0.3V to 30V
Storage Temperature Range	−65˚C to +150˚C
Junction Temperature (Note 4)	+150˚C
Soldering Information	
Infrared or Convection (20 sec.)	235˚C
Wave Soldering (10 sec.)	260˚C

Operating Ratings (Note 1)

Supply Voltage, V_S	5V to 30V
Junction Temperature Range (Note 4)	−40˚C to +85˚C
Package Thermal Resistance (Note 4)	
SOT23-5	168˚C/W

Electrical Characteristics

Unless otherwise specified, all limits guaranteed for at T_J = 25˚C, V_S = 5V. **Boldface** limits apply at temperature extremes.

Symbol	Parameter	Conditions	Min (Note 6)	Typ (Note 5)	Max (Note 6)	Units
V_{OS}	Input Offset Voltage	V_S = 5V to 30V, V_O = 1.4V, V_{CM} = 0V		2	7 **10**	mV
I_{OS}	Input Offset Current	V_O = 1.4V, V_{CM} = 0V		1.6	50 **250**	nA
I_B	Input Bias Current	V_O = 1.4V, V_{CM} = 0V		10	250 **400**	nA
I_S	Supply Current	R_L = Open, V_S = 5V		0.25	0.7	mA
		R_L = Open, V_S = 30V		0.30	2	
I_O	Output Sink Current	V_{IN^+} = 1V,V_{IN^-} = 0V, V_O = 1.5V	6	13		mA
$I_{LEAKAGE}$	Output Leakage Current	V_{IN^+} = 1V,V_{IN^-} = 0V, V_O = 5V		0.1		nA
		V_{IN^+} = 1V,V_{IN^-} = 0V, V_O = 30V		1		μA
V_{OL}	Output Voltage Low	I_O = −4mA, V_{IN^+} = 0V,V_{IN^-} = 1V		180	400 **700**	mV
V_{CM}	Common-Mode Input Voltage Range	V_S = 5V to 30V (Note 7)	V_S - 1.5V **V_S - 2V**		0 **0**	V

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Electrical Characteristics

Unless otherwise specified, all limits guaranteed for at T_J = 25˚C, V_S = 5V. **Boldface** limits apply at temperature extremes. (Continued)

Symbol	Parameter	Conditions	Min (Note 6)	Typ (Note 5)	Max (Note 6)	Units
A_V	Voltage Gain	V_S = 15V, V_O = 1.4V to 11.4V, R_L > = 15kΩ connected to V_S		120		V/mV
t_{PHL}	Propagation Delay (High to Low)	Input Overdrive = 5mV R_L = 5.1kΩ connected to 5V, C_L = 15pF		900		ns
		Input Overdrive = 50mV R_L = 5.1kΩ connected to 5V, C_L = 15pF		250		
t_{PLH}	Propagation Delay (Low to High)	Input Overdrive = 5mV R_L = 5.1kΩ connected to 5V, C_L = 15pF		940		ns
		Input Overdrive = 50mV R_L = 5.1kΩ connected to 5V, C_L = 15pF		440		ns

Note 1: Absolute Maximum Ratings indicate limits beyond which damage to the device may occur. Operating Ratings indicate conditions for which the device is intended to be functional, but specific performance is not guaranteed. For guaranteed specifications and the test conditions, see the Electrical Characteristics.

Note 2: Human body model, 1.5kΩ in series with 100pF.

Note 3: Machine model, 0Ω in series with 200pF.

Note 4: The maximum power dissipation is a function of $T_{J(MAX)}$, θ_{JA} , and T_A. The maximum allowable power dissipation at any ambient temperature is $P_D = (T_{J(MAX)} - T_A)/\theta_{JA}$. All numbers apply for packages soldered directly onto a PC board.

Note 5: Typical values represent the most likely parametric norm.

Note 6: All limits are guaranteed by testing or statistical analysis.

Note 7: The input common-mode voltage of either input should not be permitted to go below the negative rail by more than 0.3V. The upper end of the common-mode voltage range is V_S - 1.5V at 25˚C.

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